Yingfangzhong Sun

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Portfolio: www.yingfangzhong.com **Location:** Milan, Italy

Education

Politecnico di Milano, MSc in Product-Service System Design, Italy Sept 2024 – Present

- GPA: 27/30
- Relevant coursework: Software Engineering, Virtual and Physical Prototyping, Artificial Neural Networks and Deep Learning, UX Design, Data and Service

Beijing Jiaotong University, BA in Digital Media Arts, China Sept 2020 – June 2024

- GPA: 3.72/4.0 (Rank 1st)
- Relevant coursework: Graphic Programming, Design Interactions Studio, Design Interactions

Lancaster University, BA in Design Interaction Sept 2020 – June 2024

- GPA: 3.69/4.0 (Rank 1st), First Class Honours
- Dual degree programme in collaboration with Beijing Jiaotong University

Research Interests

Human–AI Interaction, Immersive Computing (AR/VR), Spatial Interaction, and Multimodal Interfaces

Publications

- **Sun, Y.**, Sandor, C., & Filonik, D. (2026). Interactive Exploration of Camera Distortions using 3D Gaussian Splats in VR. *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct '26)*. Demonstration.
- Meng, Y.[†], Ye, L.[†], **Sun, Y.**[†], Liu, B., Lu, H., & Ding, X. (2026). "Everyone Says Them": Deception Typologies, Probabilistic Trust, and Grassroots Safety Knowledge Among Gay Dating App Users in China. *ACM CSCW '26 Companion*. Poster. doi:10.1145/3785651.3831521.
- Meng, Y.[†], Fan, G.[†], Liu, B., **Sun, Y.**, Chen, R., & Mi, H. (2026). Engagement Is Not Transfer: A Withdrawal Study of a Consumer Social Robot with Autistic Children at Home. *ACM Interaction Design and Children (IDC '26)*, 755–783. doi:10.1145/3773077.3806118.

Manuscript in Preparation

- **Sun, Y.**[†], Fu, U. G.[†], Zhu, R., Lai, H. S., Meng, L., Zhang, Q., Liang, Z., & Zollmann, S. TactiCastVR: Exploring Lightweight Mediation for First-Person Soccer Tactic Sensemaking.

  The system supports first-person soccer tactical analysis in VR through formation cues, event indicators, viewpoint recommendations, and temporal traces. A controlled study ($N = 45$) found improved tactical understanding and reduced workload compared with unassisted VR.

[†] Equal contribution.

Research Experience

Max Planck Institute for Intelligent Systems (MPI-IS), Germany July 2026 – Sept 2026

Research Intern (RIGI 2026), supervised by Buse Aktaş, Felix Grüninger, and Ksenia Keplinger

Project: Embodied Work: Responsive Desk Objects for Workplace Wellbeing and Performance

- Investigated how subtle mechanical biofeedback embedded in everyday desk objects could support interoceptive awareness and potential emotion regulation, exploring physiological signals including respiration, heart rate and variability (HR/HRV), electrodermal activity (EDA), and muscle tension.
- Developed modular physical feedback prototypes incorporating pneumatic actuation, granular jamming, servo-driven movement, and vibration to explore embodied representations of physiological signals.
- Designed and conducted co-design workshops ($N = 13$), using body and workspace mapping, physical interaction samples, and iterative concept development to examine preferences for feedback placement, modalities, and interaction qualities.

Université Paris-Saclay, LISN, ARAI Team, France May 2026 – July 2026
Research Intern, supervised by Christian Sandor and Daniel Filonik
Project: Interactive Exploration of Camera Distortions using 3D Gaussian Splats in VR (Demo video)

- Investigated how to extend the visible spatial context of 3D Gaussian Splatting scenes beyond conventional perspective projection, enabling interactive exploration of wide-FOV and fisheye views in stereo VR.
- Implemented cubemap resampling and direct Gaussian projection in Unity, using local Jacobians to transform projected covariance for fisheye rendering. Demonstrated the system at IEEE ISMAR 2026.

Hefei University of Technology, MAC Lab, China Dec 2025 – Mar 2026
Researcher, supervised by Xiao Sun
Project: Vision-to-Tactile Learning and Multimodal Haptic Rendering for AR/VR

- Investigated vision-to-tactile learning to infer material properties and tactile representations from RGB images, aiming to enable visually informed haptic feedback in immersive environments.
- Developed a supervised property prediction baseline in PyTorch and a complementary pipeline for GelSight tactile image generation, with a proposed learning strategy for aligning visual and tactile representations.

Politecnico di Milano, HINT Lab, Italy Feb 2025 – June 2025
Research Intern, supervised by Maristella Matera and Ludovica Piro
Project: Intelligent Agent for Accessible Web Navigation

- Developed a voice navigation agent for blind users and people with low vision, combining speech interaction, browser automation, and accessibility metadata to support web browsing tasks.
- Helped design six interactive prototypes to compare agent suggestions and navigation approaches, and contributed to usability evaluations that informed improvements to accessible browsing.

Industry Experience

Kuaishou (Kwai), AI Agent Development Intern, China June 2025 – Sept 2025

- Developed LLM agent workflows for creative production using Dify and ComfyUI, integrating diffusion-based image generation with Python backend services and React interfaces.

Awards and Honors

Academic Dean's List (Top 10%), Lancaster University 2023 – 2024
National Third Prize, International User Experience Innovation Competition 2023
Second-Class Outstanding Scholarship, Beijing Jiaotong University (Top 10%) 2022 – 2023
Second-Class Outstanding Scholarship, Beijing Jiaotong University (Top 10%) 2021 – 2022
First-Class Outstanding Scholarship, Beijing Jiaotong University (Top 5%) 2020 – 2021

Technical Skills

Programming & Development: Python, C++, C#, C, JavaScript/TypeScript, SQL, React, FastAPI, Git, Docker
AI Frameworks & Tools: PyTorch, Hugging Face, Dify, LangGraph, ComfyUI, ControlNet
XR & Interactive Systems: Unity, Unreal Engine, OpenXR, Arduino, ESP32
Data Analysis: MATLAB, Tableau
Design & 3D Modeling: Figma, Blender, Rhino, SolidWorks, KeyShot, Adobe Creative Suite
Languages: Mandarin (Native), English (C1, IELTS Academic 7.5), Italian (B2)